QP CODE: 24900018

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION

NOVEMBER 2024

First Semester Core Course - MG1CCRBBA102 BUSINESS STATISTICS AND LOGIC

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 70 Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

General Instruction:

- *A liberal valuation can be followed*
- *For theory questions, appropriate marks can be given if the answers display the students understanding of the concept.*
- *For problem-based questions, proportionate marks can be given for steps.*

Part A

Short Answer Type Questions.

Answer any 5 out of 8 questions. Each question carries 2 marks

1. **What is a frequency polygon? [K] [1, 2]**
   - A frequency polygon is a graphical representation of a frequency distribution.
   - It is formed by plotting the midpoints of class intervals on the x-axis and their corresponding frequencies on the y-axis, then connecting the points with straight lines.
   - A diagram of frequency polygon can also be given marks
2. **Which average is suitable in the case of open end distributions? Why? [U] [1]**
   - Median (1 mark)
   - In open ended distribution, the mid value can be estimated more easily without needing the exact values of extreme categories. (1 mark)
3. **For a moderately skewed distribution, Mode and Median are respectively 65 and 68. Find Mean [A] [4]**

   Mode = 3 Median – 2 Mean

   Mean = (3Median - Mode)/2= (3*68 - 65)/2= (204 – 65) = 69.5

4. **What are the different types of measures of dispersion? [K] [1]**

   The main types are range, variance, standard deviation, and coefficient of variation.

   Note: explanation of the types is not required. Naming the types is enough.

5. **Define correlation. What does it signify in the context of statistical analysis? [U] [5]**

   - Correlation is a statistical measure that describes the strength and direction of a relationship between two variables. (1 mark)
   - It indicates how changes in one variable correspond with changes in another. (1 mark)

6. **Define probable error. If the coefficient of correlation between two variables is less than its probable error, how will interpret it in a business context. [U] [5]**

   - Probable error is a measure of the spread of data, which indicates the range within which the true value is likely to fall with a certain level of probability.

   Probable Error = $0.674 \times 1-r^2 /\sqrt{N}$

   where, r = correlation coefficient of 'n' pairs of observations for any random sample and N = Total number of observations.

   or

   Probable error (P.E.) is the value which is added or subtracted from the coefficient of correlation **(r)** to get the upper limit and the lower limit respectively, within which the value of the correlation expectedly lies.

   Probable Error = $0.674 \times 1-r^2 /\sqrt{N}$

   (1 marks)

   - There is no correlation between the variables if the value of 'r' is less than P.E. This shows that the coefficient of correlation is not at all significant. (1 marks)

7. **A call center receives an average of 12 calls per hour. What type of distribution would be appropriate to model the probability of receiving exactly 15 calls in the next hour? [A] [3]**

   Poisson distribution

   Note: No explanation required for the answer

8. **If the code for "WATER" is "XBSFS," what is the code for "FIRE"? [An] [2]**

   The code for "FIRE" is "**GJSF**."

   The code for "FIRE" would be derived using the same pattern as "WATER" to "XBSFS." For this, each letter shifts forward by 1 position in the alphabet:

   F → G; I → J; R → S; E → F

   Note: Full marks can be given for giving the correct answer without the explanation.

Part B

Short Essay Type Question
Answer any 4 out of 6 questions
Each Question carries 5 marks

9. What is classification? What are the different types of classification? [K] [1, 2]

The process of arranging data in groups or classes according to resemblances and similarities is technically called classification. By classification we try to form groups of homogeneous character. Thus, in classification of data units having common characteristics are placed in one class and the whole data are thus divided into a number of classes. *(1 mark for basic understanding of the concept)*

Methods of Classification (*5 marks)*

Classification is done on the basis of characteristics possessed by the items.

The characteristics can be of two kinds, measurable and non- measurable.

Height, weight, production etc. are measurable while sex, martial status, education etc. are not measurable. Measurable characteristics are known as variables and non - measurable characteristics are known as attributes.

Hence we have two kinds of classification.

i. Classification according to attributes.
ii. Classification according to variables.

i. Classification according to attributes (qualitative classification)

Under this method the data are classified on the basis of qualitative characteristics known as attributes. For example, literacy, unemployment etc. are attributes. They are called descriptive characteristics. They are not measurable. Their presence of absence can only be known.

Classification according to attributes may be

(a) Simple or
(b) Manifold.

In simple classification the data are divided on the basis of only one attribute. For example, population of India can be divided on the basis of sex as male and female. So, when the classification is done on the basis of one attribute there are two groups.

In manifold classification the data are divided on the basis of more than one attribute. For example, when population of India is divided on the basis of sex and literacy, there are four groups (1) Male literate (2) Male illiterate (3) Female literate (4) Female illiterate.

ii) Classification according to variables

Under this method the data are classified on the basis of quantitative characteristics such as age, height, weight etc. These characteristics are capable of direct measurement. So their different values can be known. Classification according to variables is also known as classification according to class intervals.

Height (in c.m): 135-140 140-145 145-150 150-155.155-160 160-165

Such a distribution is also known as frequency distribution.

Other classifications: Geographical classification and Chronological classification.

Geographical classification: When the data are classified according to geographical differences, we have geographical classification. Here information relating to different places are shown. For example, population of India can be shown statewise. This is a geographical classification.

Chronological classification: When data are collected for a number of periods and they are arranged chronologically we have chrono- logical classification. For example, population of India may be collected for a number of years and shown yearwise.

Note: Any other forms/ types of classification can also be considered

10. Differentiate between a bar graph and a histogram. Provide examples of situations where each would be more suitable. [U] [1, 2]

Comparison between Bar diagram and Histogram: *(3 marks)*

- Bar diagrams and Histograms both consist of bars (ie rectangles). But they differ in many respects:
- A bar diagram is a diagram, while histogram is a graph.
- In bar diagram only height (or length) of the bar is considered. Width of the bars should be the same. But in histogram area is considered, therefore both length and breadth matter.
- Other differences include:

	Bar Graph	Histogram
Definition	A bar graph represents categorical data using rectangular bars where the length is proportional to the value.	A histogram represents numerical data by grouping values into intervals (bins) and showing frequency.
Type of Data	Categorical or discrete data. Examples: fruits, colors, countries, etc.	Continuous numerical data. Examples: heights, weights, test scores, etc.
Gaps Between Bars	Bars are separated by gaps to show distinct categories.	Bars are adjacent with no gaps to represent continuous data.
Axis Labels	Categories are labeled on one axis (usually the x-axis).	Numerical intervals (bins) are labeled on one axis.
Bar Width	Bar width is uniform but arbitrary (only represents a category).	Bar width corresponds to the interval width and affects the frequency representation.
Purpose	To compare different categories or groups.	To show the distribution of data across intervals.

Example: (2 marks)

Bar Graph is used for Comparing the population of different cities or sales of different products. Histogram can be used for analysing the distribution of ages in a population or marks scored in an exam

Note: Diagrams to be given additional marks than just theory

11. Department M has a mean salary of $50,000 with 25 employees, while Department N has a mean salary of $60,000 with 15 employees. What is the combined mean salary for both departments? [A] [3,4]

$$\text{Combined Mean} = \frac{N_1\bar{X}_1 + N_2\bar{X}_2}{N_1+N_2}$$

$$\text{Combined Mean Salary} = \frac{\text{(Salary of Department M + Salary of Dept Dept N)}}{\text{Number of Employees in Dept M + Number of Employees in Dept N}}.$$

Total salaries for Department M = 50,000×25= $1,250,000

Total salaries for Department N = 60,000×15= $900,000

Total Salaries = 1,250,000 + 900,000 = $2,150,000

Total Employees = 25+15 =40

Combined Mean Salary = 2,150,000/40 = $53,750

12. An article manufactured by a company consists of two parts A and B. In the process of manufacture of Part A ,9 out of 100 are likely to be defective. Similarly, 5 out of 100 are likely to be defective in the process of manufacture of part B. Calculate the probability that the assembled part will not be defective. [A] [3]

- Let the probability of part A being defective = 9/100
- and the probability of part B being defective = 5/100
- Probability of part A not being defective = 1−(9/100) = 91/100
- Probability of part B not being defective = 1−(5/100) = 95/100.
- The probability that the assembled part will not be defective
  = (91/100) × (95/100) = 0.8645

13. Explain how to calculate range when the given data is continuous. Also find range and coefficient of range of the following series. [A] [4]

Marks	20-30	30-40	40-50	50-60	60-70
No of Students	8	12	20	7	3

- Calculating Range for Continuous Data: (4 marks)

Range for continuous series
= Upper boundary of the highest class − Lower boundary of the lowest class
Highest class is 60-70
Upper boundary of the highest class = 70
Lowest class is 20-30
Lower boundary = 20
Range = Upper boundary of the highest class − Lower boundary of the lowest class
= 70−20 =50

- Coefficient of Range: (1 marks)

$$\text{Coefficient of Range} = \frac{\text{Range}}{(\text{Upper boundary of the highest class} + \text{Lower boundary of the lowest class})}$$

$$= 50 / (70+20) = 50/90 = 0.5556$$

14. A candidate obtained the following percentage of marks. English: 60, Hindi: 75, Mathematics: 63, Physics: 60, Chemistry: 55. Find the weighted mean if weights are 1,1,2,3,3 respectively allotted to subjects. [A] [4]

$$\text{Weighted Mean} = \sum(x_i * w_i) / \sum w_i$$

Where x_i are the values and w_i are the weights.

Marks for English (X_1) = 60 and Weight for English (W_1) = 1

Marks for Hindi (X_2) = 75 and Weight for Hindi (W_2) = 1

Marks for Mathematics (X_3) = 63 and Weight for Weight (W_3) = 2

Marks for Physics (X_4) is 60 and Weight (W_4) = 3

Marks for Chemistry (X_5) is 55 and Weight (W_5) = 3

Weighted Mean = (60×1) + (75×1) + (63×2) + (60×3) + (55×3) / (1+1+2+3+3)

=60+75+126+180+165 / 10 = 606/10 = 60.6

Part C

Essay Type Questions

Answer any 2 out of 4 questions. Each question carries 20 marks.

15. Demonstrate the process of calculating the median in each of the following cases: 1) Raw data, 2) Ungrouped data, and 3) Grouped data.
Provide an example for each scenario to illustrate the steps involved.

Solution:

Note: 10 marks for the explanation and 10 marks for the example. The examples need not be accurate.

1. Median for Raw Data

Raw data refers to unorganized data points. Steps to calculate median for raw data is:

i. Arrange the data in ascending order
ii. If the number of observation is odd, Median is the middle value in series

Example: Let's find the median of the following raw data:
9,3,7,5,6
Step 1: Arrange the data in ascending order:
3,5,6,7,9
Step 2: Identify the middle value: here, Number of observations (n) = 5 (odd).
Median position = (n+1) / 2= (5+1) / 2 =3
The 3rd value is 6.

2. Median for Ungrouped Data

Ungrouped data consists of data presented as a frequency distribution.

Example:

Find the median for the following frequency distribution:

Data (x)	3	5	7	9
Frequency (f)	2	3	4	1

Steps:

1. Calculate the cumulative frequency (CF):
2. Determine the total number of observations:
3. Find the median position:
4. Locate the median class:

Data (x)	3	5	7	9
Frequency (f)	2	3	4	1
Cumulative Frequency (CF)	2	5	9	10

Median: 7

3) Median for Grouped Data

Grouped data is organized into classes with corresponding frequencies.

Steps:

1. Calculate the cumulative frequency (CF):

Class Interval	0-10	20-30	20-30	30-40
Frequency (f)	5	8	12	6
Cumulative Frequency (CF)	5	13	25	31

2. Determine the total number of observations:
   - N=∑f=31.
3. Find the median class:
   - Median position = N/2=31/2=15.5
   - The cumulative frequency just exceeding 15.5 , corresponding to the class interval 20−30
   - Median class = 20−30
4. Apply the median formula:

$$\text{Median} = L + \left(\frac{\frac{n}{2} - CF}{f} \right) \cdot h$$

16. Find mean deviation about the arithmetic mean for the following frequency distribution of marks of 60 college students.

Marks : 0 - 10 10 - 20 20 - 30 30 - 40 40 - 50 50 - 60 60 - 70
Frequency : 4 6 10 20 10 6 4
[A] [3]

Solution:

Class	**Frequency (f)**	**Midpoint (*x*)**	**f * *x***	**|*x* - Mean|**	**f * |*x* - Mean|**
0-10	4	5	20	30	120
20-30	6	15	90	20	120
20-30	10	25	250	10	100
30-40	20	35	700	0	0
40-50	10	45	450	10	100
50-60	6	55	330	20	120
60-70	4	65	260	30	120
	∑f=60		**∑(f * *x*)** = 2100		**∑(f * |*x* - Mean|)** =680

$$\text{Arithmetic Mean} \quad \bar{x} = \frac{\sum f \cdot x}{\sum f}$$

Where,

$$\text{Mid-point of class } (x) = \frac{\text{Lower Limit} + \text{Upper Limit}}{2}$$

From the table above, Arithmetic mean = 2100/60 = 35

$$\text{Mean Deviation} = \frac{\sum f \cdot |x - \bar{x}|}{\sum f}$$

Mean Deviation = 680/60 = 11.33

17. a) What is the role of the standard error in regression analysis? B) What are the key limitations associated with regression analysis? [K] [5]

Solution:

(a) The Role of the Standard Error in Regression Analysis *(10 marks)*

The Standard Error (SE) plays a crucial role in evaluating the reliability and precision of regression estimates.

- Precision of Coefficients: The standard error measures the variability or dispersion of the regression coefficients. Smaller SE values indicate that the estimates are more precise.
- Significance Testing: SE is used to calculate the t-statistic for each regression coefficient ($t = \beta/SE$), which helps test the null hypothesis (that the coefficient is zero). A smaller SE increases the likelihood of rejecting the null hypothesis, indicating a statistically significant relationship.
- Goodness of Fit: The standard error of the regression measures the average distance that the observed values fall from the regression line. Smaller values indicate better model fit.
- Confidence Intervals: SE is used to construct confidence intervals for regression coefficients. This helps assess the range within which the true coefficient is likely to lie.
- Model Comparison: SE can be used to compare the fit of different models, especially when assessing residual errors.

(b) Key Limitations of Regression Analysis

- Assumptions: Regression analysis relies on assumptions like linearity, homoscedasticity (constant variance of errors), independence of errors, and normality of residuals. Violations of these assumptions can lead to biased or invalid results.
- Causation vs. Correlation: Regression only identifies relationships, not causation. Misinterpreting a strong correlation as causation can lead to flawed conclusions.
- Overfitting: Including too many predictors in a regression model can lead to overfitting, where the model performs well on the training data but poorly on new data.
- Omitted Variable Bias: If a relevant variable is excluded, the model can produce biased and inconsistent estimates, distorting the results.

- Multicollinearity: When independent variables are highly correlated, it becomes challenging to determine the individual effect of each variable, leading to unreliable coefficient estimates.
- Outliers and Influential Points: Extreme values or influential data points can disproportionately affect the regression line and coefficients, reducing the robustness of the model.
- Limited to Linear Relationships: Standard regression techniques are suited for linear relationships. For nonlinear relationships, more complex models are required.
- Sample Size: Small sample sizes can lead to high variability in coefficient estimates and reduced statistical power.

18. From the following data, obtain the two regression equations
Sales : 91 97 108 121 67 124 51 73 111 57
Purchases: 71 75 69 97 70 91 39 61 80 47 [A] [5]

Solution:

X (Sales)	Y (Purchases)	XY	X^2	Y^2
91	71	6461	8281	5041
97	75	7275	9409	5625
108	69	7452	11664	4761
121	97	11737	14641	9409
67	70	4690	4489	4900
124	91	11284	15376	8281
51	39	1989	2601	1521
73	61	4453	5329	3721
111	80	8880	12321	6400
57	47	2679	3249	2209
900	**700**	**66900**	**87360**	**51868**

- For regression of Sales on Purchases (Y on X): $Y=a_{YX}+b_1X$

Slope $b_1 = [n(\Sigma XY) - (\Sigma X)(\Sigma Y)] \div [n(\Sigma X^2) - (\Sigma X)^2]$

Intercept $a_{YX} = [\Sigma Y - b1(\Sigma X)] \div n$

Substitute the values:

- $n=10; \Sigma XY=66900; \Sigma X=900; \Sigma Y=700; \Sigma X^2=87360$

$b_1 = [(10)(66900) - (900)(700)] / [(10)(87360) - (900)^2] = 0.613$

$a_{YX} = [700 - (0.613 \text{ X } 900)] / 10 = 14.81$

Thus, the regression equation for Purchases on Sales (Y on X): $Y=14.811+0.613 \cdot X$

- For regression of Purchases on Sales (X on Y): $X=a_{XY}+b_2X$

Slope $b_2 = [n(\Sigma XY)-(\Sigma X)(\Sigma Y)] \div [n(\Sigma Y^2)-(\Sigma Y)^{2]}$

Intercept $a_{XY}= [\Sigma X-b'1(\Sigma Y)] \div n$

Substitute the values:

- $n=10$; $\Sigma XY=66900$; $\Sigma X=900$; $\Sigma Y=700$; $\Sigma Y^2=51868$

$b_2= [(10)\ (66900)\ -(900)\ (700)]\ /\ [(10)\ (51868)\ -(700)^2] = 1.36$

$a_{XY} =[900-(1.36 \times 700)]\ /\ 10 = (900-951.883)\ /10 = -5.189$

Thus, the regression equation for Sales on Purchases (X on Y): $X=-5.189+1.36 \cdot Y$

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